



• ELECTRICAL ENGINEERING • LESSON 01

Introduction to Electricity

Voltage, Current, Resistance & How Power Flows

Lesson 1 of 30 | Electrical Engineering Program | Beginner

Awet G. Nway

Founder, AYE Tech Hub | Electrical & Power Systems Engineer

- What electricity is — electrons, charge, current, and voltage
- Ohm's Law and the power equation — every engineer's starting point
- AC vs DC — how two different forms of power are used in the world
- How electricity gets from power station to your outlet (grid overview)
- Electrical safety fundamentals and career pathways

01 | What Is Electricity?

Electricity is the flow of **electric charge** — specifically, the movement of electrons through a conductor. Everything in electrical engineering traces back to this fundamental idea: when electrons are pushed through a material by a voltage source, they carry energy from one place to another, doing useful work along the way — lighting lamps, driving motors, heating elements.

Understanding electricity begins with four quantities that describe every electrical circuit. These four — **Voltage, Current, Resistance, and Power** — are related by two simple equations that every electrical engineer uses daily: Ohm's Law and the Power Equation.

Quantity	Symbol	Unit	Analogy (Water Pipe)	Measured With
Voltage (V)	V or U	Volt (V)	Water pressure — the force that pushes electrons	Voltmeter (across component)
Current (I)	I	Ampere (A)	Flow rate — how many electrons pass per second	Ammeter (in series)
Resistance (R)	R	Ohm (Ω)	Pipe narrowness — how much the material opposes flow	Ohmmeter (power off)
Power (P)	P	Watt (W)	Rate of work done — energy transferred per second	Wattmeter or $P = V \times I$
Energy (E)	E	Joule / kWh	Total water delivered — power used over a time period	Energy meter (kWh meter)

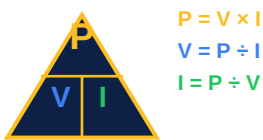
02 | Ohm's Law & the Power Equation

Georg Simon Ohm discovered in 1827 that the current through a conductor is **directly proportional to the voltage across it** and inversely proportional to its resistance. This relationship — $V = I \times R$ — is the single most used equation in all of electrical engineering. Combine it with the power equation $P = V \times I$ and you can solve the majority of circuit analysis problems.

Ohm's Law Triangle



Power Triangle

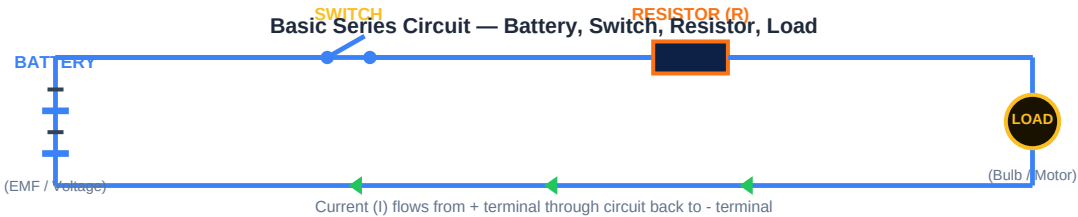


EXAMPLE — Using Ohm's Law in Practice

A 230V supply powers a 100W light bulb. How much current flows, and what is the bulb's resistance? Step 1: $I = P \div V = 100 \div 230 = 0.435 \text{ A}$. Step 2: $R = V \div I = 230 \div 0.435 = 529 \text{ }\Omega$. Result: the filament has 529 ohms of resistance and draws 0.435 amps at full brightness.

03 | The Basic Circuit — Components and Current Flow

Every electrical circuit, from a torch to a power station, contains the same fundamental elements: a **source** (battery or generator), a **path** (conducting wires), a **load** (device that uses energy), and a **control** (switch or protective device). Current flows in a continuous loop from the positive terminal of the source, through the circuit, back to the negative terminal.



04 | AC vs DC — The Two Forms of Electrical Power

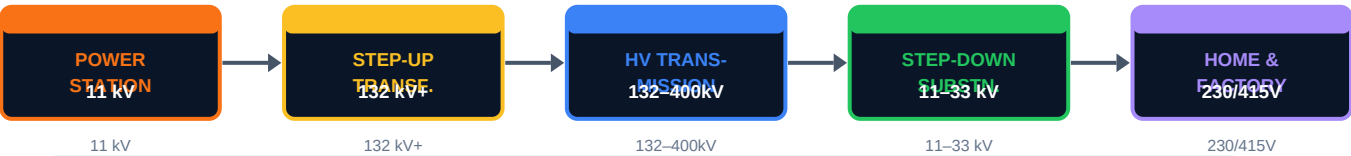
Electricity exists in two fundamentally different forms. **Direct Current (DC)** flows in one direction only. **Alternating Current (AC)** reverses direction periodically — typically 50 or 60 times per second (50/60 Hz). Both are essential in modern electrical systems, and understanding the difference is critical for every electrical engineer.

Property	DC (Direct Current)	AC (Alternating Current)
Direction of flow	Constant — always flows one way (+ to -)	Reverses direction at the supply frequency (50/60 Hz)
Voltage waveform	Flat horizontal line — constant voltage	Sinusoidal wave — varies between +peak and -peak
Generation	Batteries, solar panels, fuel cells, rectifiers	Generators, alternators — all grid power is AC
Transmission	Loses voltage over long distances — inefficient	Easily stepped up/down with transformers — ideal for grids
Typical voltages	1.5V (AA cell) → 12V (car) → 48V (telecom) → 800V (EV battery)	230V/50Hz (Europe/Africa) or 120V/60Hz (Americas)
Where used	Electronics, computers, LED lighting, EVs, solar storage	Mains power, motors, industrial machinery, HVAC systems
Key advantage	Precise control; easy to store in batteries	Can be transformed to any voltage; efficient long-distance transmission

05 | How Electricity Reaches You — The Power Grid

The electricity you use at home travels through a complex chain of equipment from power station to outlet. Understanding this chain — called the **electrical grid** — is fundamental for power engineers, building services engineers, and anyone designing electrical installations.

How Electricity Reaches You — From Power Station to Outlet



Stage	Equipment	Voltage	Why This Voltage?
Generation	Turbine-driven alternators (steam, gas, hydro, wind, nuclear)	11–25 kV	Practical voltage for large generator windings — not too high for insulation
Step-Up Substation	Large power transformers (T1)	132–400 kV	High voltage = low current = low losses in transmission cables ($P_{\text{loss}} = I^2R$)
HV Transmission	Overhead lines on pylons or underground cables	132–400 kV	Minimises energy losses over hundreds of kilometres
Grid Substation	Step-down transformers (T2), busbars, circuit breakers	33–66 kV	Reduces voltage for regional distribution networks
Distribution Network	Overhead/underground cables, pole-top transformers	11 kV	Supplies industrial and commercial premises directly at 11 kV
Service Transformer	Distribution transformer (pole-mounted or padmount)	230/415 V	Final step-down for domestic and small commercial consumers
End User	Consumer unit (distribution board), wiring, outlets	230V (1-ph) / 415V (3-ph)	Safe, standard voltage for appliances and equipment

06 | Electrical Safety Fundamentals

Electricity is silent, invisible, and instant. A current as small as **50 mA (0.05 A)** through the chest can cause **cardiac arrest**. Every electrical engineer, technician, and installer must understand and apply safety principles before touching any live equipment.

Current Level	Effect on Human Body	Risk Level
< 1 mA	No sensation — imperceptible	Safe
1–10 mA	Mild tingling or shock sensation	Low
10–20 mA	Painful — muscle contraction; may not release grip	DANGEROUS
20–100 mA	Severe muscular contractions; breathing difficulty; ventricular fibrillation possible	LETHAL ZONE
> 100 mA	Ventricular fibrillation — almost certain death without immediate defibrillation	LETHAL
> 1 A	Severe burns, tissue destruction, cardiac arrest	LETHAL

THE GOLDEN RULES — Electrical Safety

1. ISOLATE: Always switch off and lock out the supply before working. 2. TEST: Use a voltage tester to confirm the circuit is dead — never assume. 3. EARTH: Ensure all metal enclosures are properly earthed. 4. PROTECT: Use PPE (insulated gloves, safety boots), MCBs, RCDs, and fuses. 5. NEVER WORK LIVE unless specifically trained, authorised, and equipped to do so.

07 | Electrical Engineering Career Pathways

Role	Core Responsibilities	Key Skills	Avg Salary (USD/yr)
------	-----------------------	------------	---------------------

Electrical Design Engineer	Design electrical systems: power, lighting, distribution, protection for buildings and industry	AutoCAD Electrical, Revit MEP, load calc, IEE Wiring Regs / NEC	\$60,000 – \$95,000
Power Systems Engineer	Design and analyse generation, transmission, and distribution grids	Power flow analysis, protection relay coordination, PSS/E, ETAP	\$70,000 – \$110,000
Building Services Engineer	Electrical systems for buildings: LV, lighting, fire alarms, security, UPS	IEE BS 7671 / NEC, AutoCAD, Revit, energy efficiency standards	\$55,000 – \$90,000
Control & Instrumentation Engineer	Design measurement and control systems for industrial processes	PLC/SCADA, instrumentation loops, P&IDs, SIL analysis	\$65,000 – \$105,000
Solar & Renewable Engineer	Design photovoltaic and wind energy systems	PVsyst, load sizing, inverter selection, grid code	\$55,000 – \$90,000
Electrical Project Manager	Manage electrical installation projects from design to commissioning	MS Project, cost control, procurement, risk management	\$75,000 – \$120,000

Reflection:

"Look at any plug socket near you. It delivers 230V at up to 13A (in the UK) or 120V at 20A (in the US). Using $P = V \times I$, calculate the maximum power available from that outlet. Now look at the back of any appliance — does its rated power match what the outlet can provide? You have just done a basic load calculation."

Action checklist:

Next up — ELEC-002:

Community & support:

Basic Electrical Quantities — voltage, current, resistance, power, energy, and the SI units every engineer must know.

[t.me/ayetechub](#)